

SIDDHARTH CHAINI

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EDUCATION

University of Delaware

Newark, DE, USA

Ph.D. in Physics and Astronomy (*GPA: 3.96/4.0*)

Sep. 2022 – Apr. 2027 (expected)

NASA FINESST Fellow & UD Doctoral Fellow for Excellence

Advisor: Federica B. Bianco

Dissertation: *“Discovering Structure and Surprise in Time-Domain Astrophysics”*

Indian Institute of Science Education and Research (IISER) Bhopal

Bhopal, India

BS-MS (Dual Degree) in Physics (*CPI: 9.04/10*)

Aug. 2017 – May 2022

PUBLICATIONS



* marks equal contribution; author lists beyond three names are shortened to “et al. (incl. **S. Chaini**)”.

Lead & Major-Contribution Publications

- [11] N. Earl, **S. Chaini**, K. D. French, et al., “NAPTImE: A Neural-Process Framework for Rubin Alert Classification,” *ApJ*, Jul. 2026. arXiv: [2607.19236](https://arxiv.org/abs/2607.19236). *Accepted*.
- [10] **S. Chaini**, F. B. Bianco, and A. Mahabal, “Probabilistic Data-Driven Modelling of Astrophysical Transients: The Neural Process Family for Ultrafast and Class-Agnostic Light Curve Reconstruction with NightLANP,” *arXiv e-prints*, May 2026. arXiv: [2605.27527](https://arxiv.org/abs/2605.27527). *Submitted*.
Also: ICML 2026 Workshop on AI for Physics [\[OpenReview\]](#).
- [9] **S. Chaini**, F. B. Bianco, and A. Mahabal, “In Search of the Unknown Unknowns: A Multi-Metric Distance Ensemble for Out of Distribution Anomaly Detection in Astronomical Surveys,” *arXiv e-prints*, Oct. 2025. arXiv: [2510.23702](https://arxiv.org/abs/2510.23702).
- [8] B. Domínguez*, **S. Chaini***, K. Mrakovčić*, et al., “Validation of the Bond et al. (2010) SDSS-derived kinematic models for the Milky Way’s disk and halo stars with Gaia Data Release 3 proper motion and radial velocity data,” *A&A*, vol. 688, A132, Aug. 2024, Art. no. A132. DOI: [10.1051/0004-6361/202449653](https://doi.org/10.1051/0004-6361/202449653). arXiv: [2406.03541](https://arxiv.org/abs/2406.03541).
- [7] **S. Chaini**, A. Mahabal, A. Kembhavi, et al., “Light curve classification with DistClassiPy: A new distance-based classifier,” *Astronomy and Computing*, vol. 48, p. 100850, Jul. 2024, Art. no. 100850. DOI: [10.1016/j.ascom.2024.100850](https://doi.org/10.1016/j.ascom.2024.100850). arXiv: [2403.12120](https://arxiv.org/abs/2403.12120).
- [6] **S. Chaini**, A. Bagul, A. Deshpande, et al., “Photometric identification of compact galaxies, stars, and quasars using multiple neural networks,” *MNRAS*, vol. 518, no. 2, pp. 3123–3136, Jan. 2023. DOI: [10.1093/mnras/stac3336](https://doi.org/10.1093/mnras/stac3336). arXiv: [2211.08388](https://arxiv.org/abs/2211.08388).
- [5] **S. Chaini** and S. S. Kumar, “Astronomical Classification of Light Curves with an Ensemble of Gated Recurrent Units,” *arXiv e-prints*, Jun. 2020. arXiv: [2006.12333](https://arxiv.org/abs/2006.12333).

Other Co-Authored Publications

- [4] C. R. Bom, A. Mahabal, F. Bianco, et al. (incl. **S. Chaini**), “Toward decision-aware AI for LSST-scale time-domain astronomy,” *arXiv e-prints*, Jun. 2026. arXiv: [2606.05285](#). *Submitted, Nature Astronomy*. Rubin LSST Transients and Variable Stars Science Collaboration.
- [3] E. J. Honaker, J. E. Gizis, C. Aganze, et al. (incl. **S. Chaini**), “Searching for Ultracool Dwarfs in Early LSST Data Products,” *ApJ*, vol. 1004, no. 2, p. 214, Jun. 2026, Art. no. 214. DOI: [10.3847/1538-4357/ae6ce9](#). arXiv: [2604.08387](#).
- [2] C. L. Ransome, B. M. Subrayan, D. J. Sand, et al. (incl. **S. Chaini**), “First Results from the LSST Shadow Survey: The Restless Luminous Blue Variable AT2017des in the Virgo-Cluster Galaxy, NGC4532,” *ApJL*, Jun. 2026. arXiv: [2606.23784](#). *Submitted*. The Shadow Collaboration.
- [1] D. J. Sand, B. Subrayan, C. L. Ransome, et al. (incl. **S. Chaini**), “Nearby Supernova Science in the Era of High Cadence, Deep Time Domain Surveys,” 2026. *In preparation*. The Shadow Collaboration.

Software

- S. Chaini**, F. B. Bianco, and A. Mahabal, “*NightLANP*”, GitHub repository, Jun. 2026. [Online]. Available: <https://github.com/sidchaini/NightLANP/>.
- S. Chaini**, F. B. Bianco, and A. Mahabal, “*keras-neural-processes*”, GitHub repository, Nov. 2024. [Online]. Available: <https://github.com/sidchaini/keras-neural-processes/>.
- S. Chaini**, A. Mahabal, A. Kembhavi, et al., “*DistClassiPy*”, Astrophysics Source Code Library, record ascl:2403.002, Mar. 2024. [Online]. Available: <https://github.com/sidchaini/DistClassiPy/>.

TECHNICAL SKILLS

Surveys & data	Rubin/LSST (DP0–DP2, alert-stream simulations), DECam, ZTF, SDSS, <i>Gaia</i> DR3
Languages	Python, C, C++, Java, Wolfram Language, SQL, HTML, Bash, Assembly, L ^A T _E X
Libraries	PyTorch, TensorFlow/Keras, scikit-learn, Astropy, NumPy, pandas, Matplotlib, Selenium
Tools	Git, Jupyter, VS Code, Mathematica, SAOImage DS9, Aperture Photometry Tool

FELLOWSHIPS, GRANTS AND AWARDS

- **NASA FINESST** — Future Investigators in NASA Earth and Space Science and Technology 2025 – 2027
Proposal written as the Future Investigator; 24 of 456 Astrophysics proposals selected (~5%)
- **NSF Research Grant** — National Science Foundation 2025 – 2028
Co-author on the funded proposal (PI: F. B. Bianco); graduate student researcher
- **Doctoral Fellowship for Excellence**, *University of Delaware* – 2 awarded per year in Physics 2025 – 2026
- LINCC Frameworks Community Event Support, *LSST Discovery Alliance* 2026
- **Data Science Institute Fellow**, *University of Delaware* 2023 – 2027
- Vatican Observatory Scholarship, *18th Summer School* – 25 selected from ~200 applicants, 20 countries 2023
- **INSPIRE** Scholarship, *DST, Government of India* – Top 1% of ~1.2 million applicants nationally 2017 – 2022

OBSERVING EXPERIENCE

- **Dark Energy Camera (DECam)**, Cerro Tololo Inter-American Observatory – Remote observer, DECam Shadow project 2 nights

ACADEMIC SERVICE & LEADERSHIP

- **Elected Member**, LSST ISSC Executive Council Jun. 2026 – Present
Informatics and Statistics Science Collaboration — oversight of working groups, committee appointments and collaboration policies
- **Co-Chair**, LSST TVS Anomalies Subgroup Sep. 2024 – Present
Rubin LSST Transients and Variable Stars Science Collaboration — organize meetings and coordinate subgroup activities
- **Organizing Committee**, LSST is here! DP2 Hackathon Aug. 2026
Co-organized a 2-day hackathon on Rubin Observatory's LSST Data Preview 2 (DP2) data and tools
- **Instructor**, Rubin TVS Software Taskforce – Multiple collaboration-wide software workshops Aug. 2026
- **Reviewer**, workshop programme committees – ICML Workshop on AI for Physics; NeurIPS Workshop on Representations for Physical Sciences 2026
- **Elected Head**, Student Research Group, IISER Bhopal Astronomy Club Aug. 2020 – Aug. 2021
In charge of group activities and data analysis — image reduction and CCD photometry

SELECTED TALKS

- Rubin Community Workshop 2026 ([link](#)) – Fast, Probabilistic Modelling for the LSST Alert Stream Jul. 2026
- The Extragalactic Transient Universe ([link](#)) – Probabilistic Modelling of Transient Light Curves with Neural Processes Jul. 2026
- European Astronomical Society Annual Meeting – Using neural processes to model sparse light curves for LSST-like transients Jun. 2026
- American Physical Society Global Physics Summit ([link](#)) – Tackling Sparse, Multiwavelength Light Curves in Time-Domain Astronomy with Neural Processes Mar. 2026
- TDABench – The Need for a Light Curve Interpolation and Forecasting Benchmark Dataset Mar. 2026
- Quasar Bazaar Hackweek ([link](#)) — **Invited** (remote) – Distance Multi-Metric Anomaly Detection (DiMMAD) Feb. 2026
- NCRA Astronomy Seminar ([link](#)) — **Invited** – Transient hunting in the era of large-sky surveys with Neural Processes Jan. 2026
- LSST@Europe 7 ([link](#)) – Tackling LSST's sparse multiband light curves with Neural Processes Sep. 2025
- Rubin Community Workshop 2025 ([link](#)) – Hunting for Anomalies with Distance Metrics for LSST Jul. 2025
- Shaping the Future of Time-Domain Astronomy with LSST – Anomalies in the time-domain: Machine learning with distance metrics May 2025
- AI/ML Applications in Astronomy and Astrophysics — **Invited Panelist** – Pitfalls of using ChatGPT Jan. 2025
- AI/ML Applications in Astronomy and Astrophysics ([link](#)) — *Hands-on tutorial* – Leveraging Distance Metrics for Better Machine Learning Jan. 2025
- LSST TVS Colloquium ([link](#)) — **Invited** – Distance Metrics for Machine Learning in Time-Domain Astronomy with DistClassiPy Oct. 2024
- LSST@Europe 6 ([link](#)) – Leveraging Distance Metrics for Machine Learning in Astronomy: DistClassiPy Sep. 2024
- Rubin Community Workshop 2024 ([link](#)) – DistClassiPy: Distance-Based Techniques for Classification and Anomaly Detection in Light Curves Jul. 2024

SELECTED POSTERS & WORKSHOPS

- European Astronomical Society Annual Meeting 2026 ([link](#)) – What physics can we find in the data-driven manifolds of supernovae spectra? Jul. 2026
- IAU Symposium 406: Future landscape of astrophysical transients ([link](#)) – Attentive Neural Processes model LSST-like transient light curves more accurately — and 100,000× faster — than Gaussian Processes May 2026
- NeurIPS 2025, *Machine Learning and the Physical Sciences* ([link](#)) – DiMMAD: a multi-metric distance ensemble for out-of-distribution anomaly detection Dec. 2025
- Rare Gems in Big Data ([link](#)) – DistClassiPy: A New Distance-Based Classifier May 2024
- APS Mid-Atlantic Section Annual Meeting ([link](#)) – Light curve classification using distance metrics Nov. 2023
- 40th Meeting of the Astronomical Society of India ([link](#)) – Photometric classification of compact galaxies, stars and quasars using multiple neural networks Mar. 2022
- Institut d’Astrophysique de Paris Colloquium ([link](#)) – Photometric classification of faint & compact galaxies, stars and quasars using multiple neural nets Oct. 2021

MENTORING

- Co-mentored three Lincoln University undergraduates May 2026 – Aug. 2026
Abdourahim Sylla, Agustina Lorda and Adeoluwa Afolabi
- Co-mentored a UD undergraduate NASA Space Grant Fellow – Ibrahim Wilson Jul. 2023 – Aug. 2024

TEACHING EXPERIENCE

- **Lesson Instructor**, Rubin LSST TVS Science Collaboration Jul. 2024 – Present
Intermediate Python for Astronomical Software Development; Research Software Engineering Mini Course
- **Teaching Assistant**, University of Delaware – PHYS 207/208/227, SCEN 101 Sep. 2022 – Aug. 2024
- **Teaching Assistant**, IISER Bhopal – ECS 102: Introduction to Programming Jan. 2019 – May 2019

OUTREACH

- **Rubin First Look**, University of Delaware — co-led a public viewing party for Rubin Observatory’s first images, and presented LSST’s goals and science to the public.
- **Astronomy Club**, IISER Bhopal — organized telescope viewing nights and computational-astronomy programming tutorials for the campus community.
- **Outreach talks**, tutoring centres in Thane, India — science talks to students in grades 8–10 to spark interest in astrophysics.
- **Media coverage** — *UDaily*: “[NASA’s next generation of scientists](#)” (Oct. 2025), “[Opening a new window into the Universe](#)” (Jun. 2025); *Franciscan Media*: “[Vatican hosts astronomers of tomorrow](#)” (Jun. 2023, also on [YouTube](#)).

REFERENCES

- **Federica B. Bianco** (University of Delaware) — *PhD advisor* fbianco@udel.edu
- **Ashish Mahabal** (Caltech) — *MS thesis co-advisor* aam@astro.caltech.edu
- **Željko Ivezić** (Rubin Observatory/UW) ivezic@uw.edu